Project Proposal Presentation (3/2)

- Upload to Assignments > Term Project > Term Project Proposal
  - 5% of your project grade
  - Teams are required to present a 5-10 minute proposal to introduce their project to the class. The presentation should be in PPT or PDF format and include at least 5 slides covering the following:
    - **Team Name and Project Overview**: Briefly introduce the team and outline the project objectives.
    - **AWS Services Overview**: List the AWS services you plan to use, with a brief description of how each will support project goals.
    - **High-Level Data Flow Diagram:** Present a diagram showing the interaction of AWS services and the primary data flow within the project.
  - Only slides with substantive content count toward the minimum; placeholder slides like "Overview" or "Questions" do not.

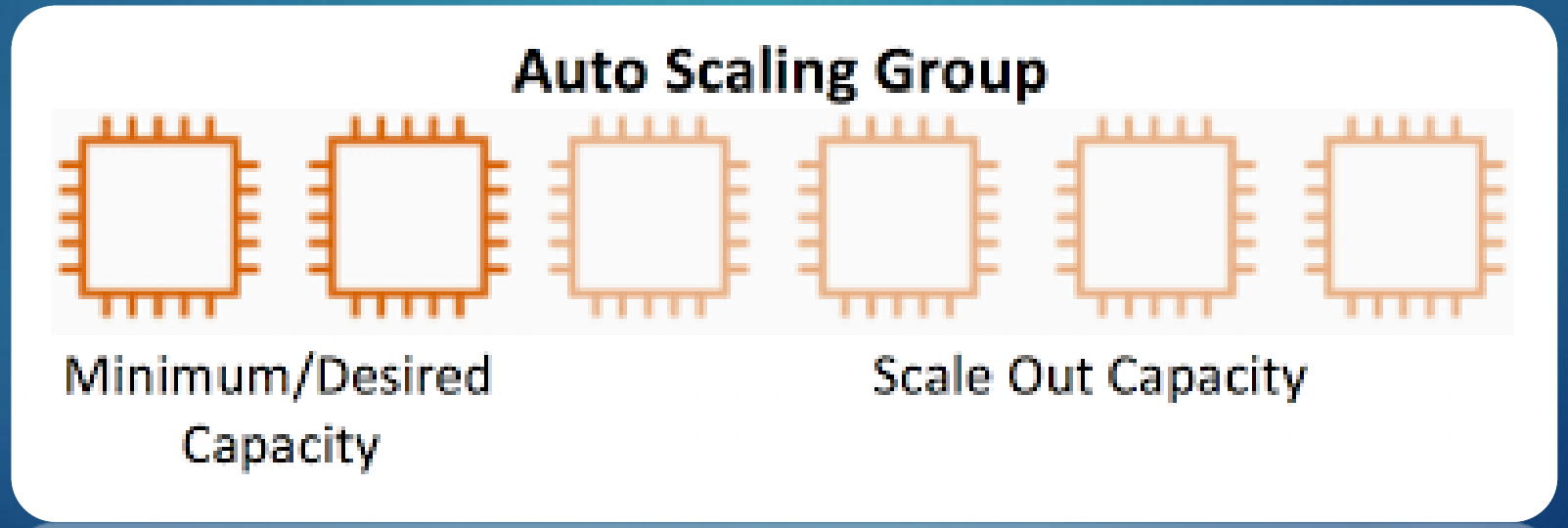
Project Check-in (1 of 3)

- Check-in is worth 5 points
- Details on myCourses under Key Links > Project Check-in
- Sign-ups next week via Calendy
  - I will post a note in Slack and myCourses

- From the previous exercise you provided with the AWS Cost Calculator, you've been asked to investigate opportunities to use Auto-Scaling in order to reduce the costs from your initial estimate
- You've been tasked to identify the following:
  - What Auto-Scaling usage pattern applies to your scenario?
    - What type of factors impact this usage pattern?
  - What type of Auto-Scaling plan would you use to scale in or out?
  - What would you Auto-Scale on (CPU, memory, other factors)?

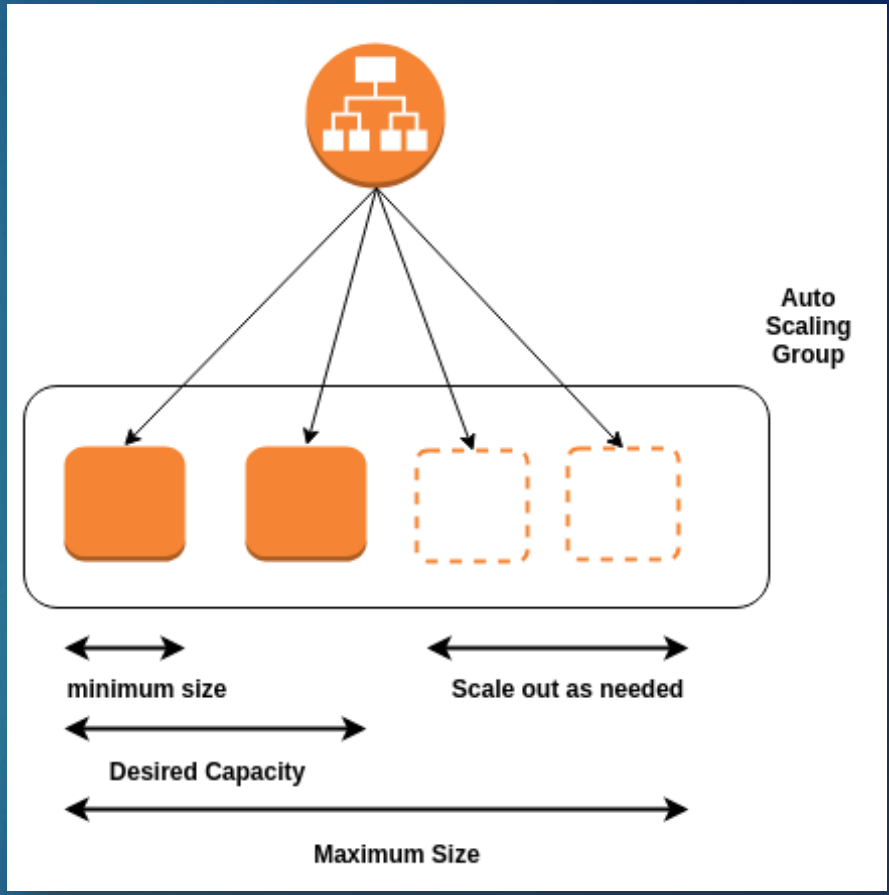
RECAP

- Auto-Scaling is a cloud computing feature that enables organizations to scale cloud services such as virtual machines up or down automatically, based on defined situations (e.g. increased usage)
- The overall benefit of Auto-Scaling is that it eliminates the need to respond manually in real-time to traffic spikes that merit new resources and instances by automatically changing the active number of servers



RECAP

1. Launch template
   - Contains the AMI, server type, etc. that Auto-Scaling groups use to launch the EC2 instances
2. Auto-Scaling group
   - A collection of EC2 instances that are treated as a logical grouping for the purposes of automatic scaling and management
3. Auto-Scale plan
   - Defines "when and how" to scale the Auto-Scaling group. Includes:
     - Health check time (how quickly to adjust)
     - Desired #, Min/Max #



- By associating a load balancer to an Auto-Scaling group, applications can quickly scale-up to handle the increased demand with no downtime

Containers in the Cloud

Engineering Cloud Software Systems

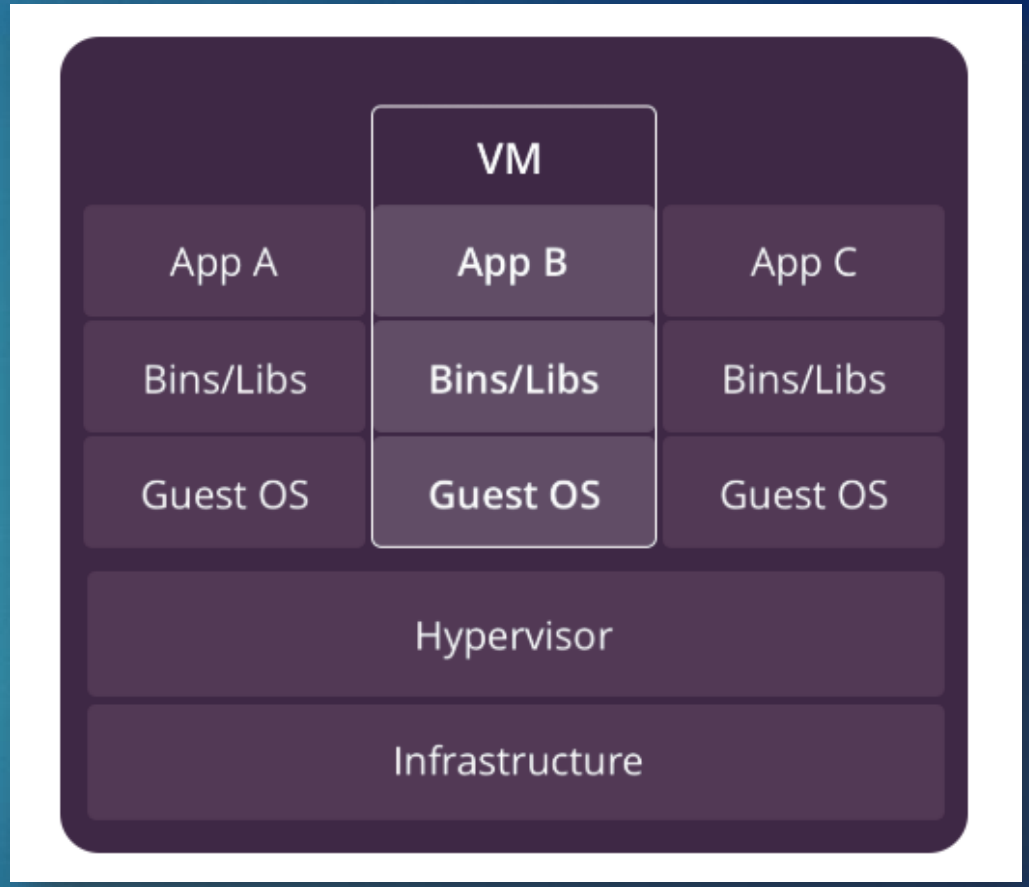
Department of Software Engineering
Rochester Institute of Technology

EC2 Instances and Auto-Scaling…

- One drawback to starting up a new EC2 instance is it can several seconds or even minutes before it's ready
  - They can consume a lot of resources and may not be required if you need to do something small and simple (e.g. read a message from a queue and write to a database)
- What if we needed something more lightweight using less resources and is more cost efficient?
  - Instead of taking minutes to start up, could we start up faster (e.g. milliseconds)?
- In this scenario, virtual machines may not be the right choice

Virtual Machines (VM) – Recap

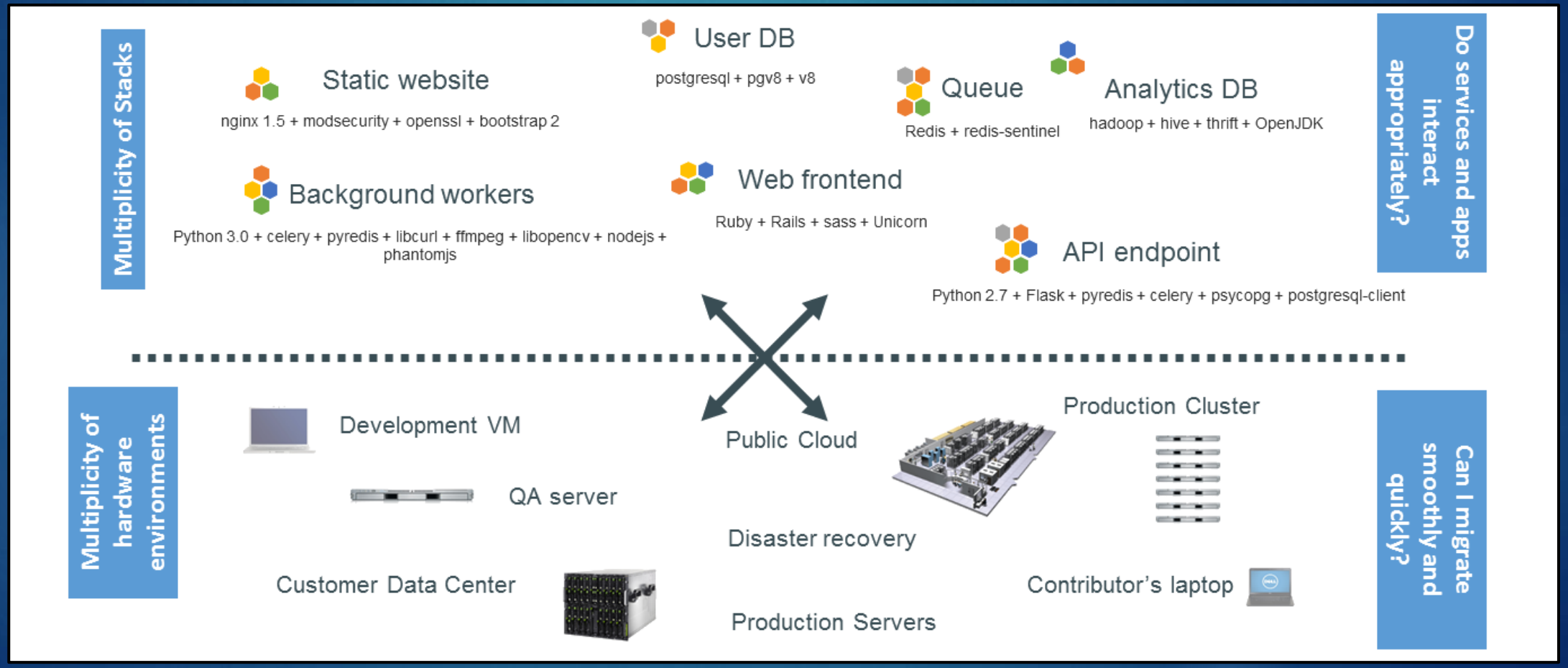
- A hypervisor creates and runs VMs. It's sits between the hardware and the VM to virtualize the server
- Within each VM runs one or more unique guest operating systems
- VMs with different operating systems (e.g. Linux, Windows) can run on the same physical server
- Each VM has its own binaries, libraries, and applications that it services
- A VM could be several gigabytes in size



Source: https://www.backblaze.com/blog/vm-vs-containers/

Consider this Problem

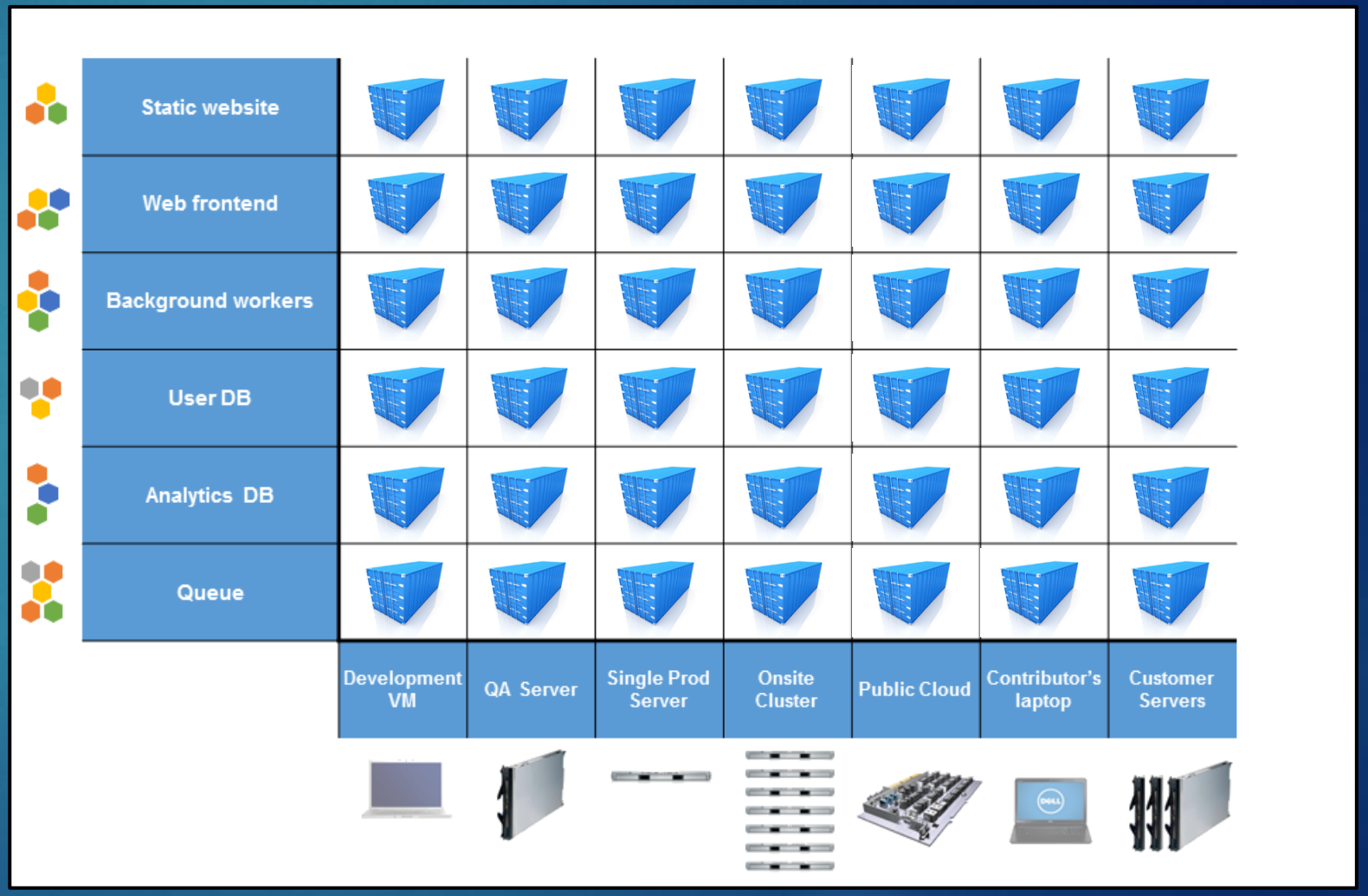
- When software (top) is deployed to a hardware environment (bottom), how many permutations are there?



Source: http://pointful.github.io/docker-intro/

Consider this Problem

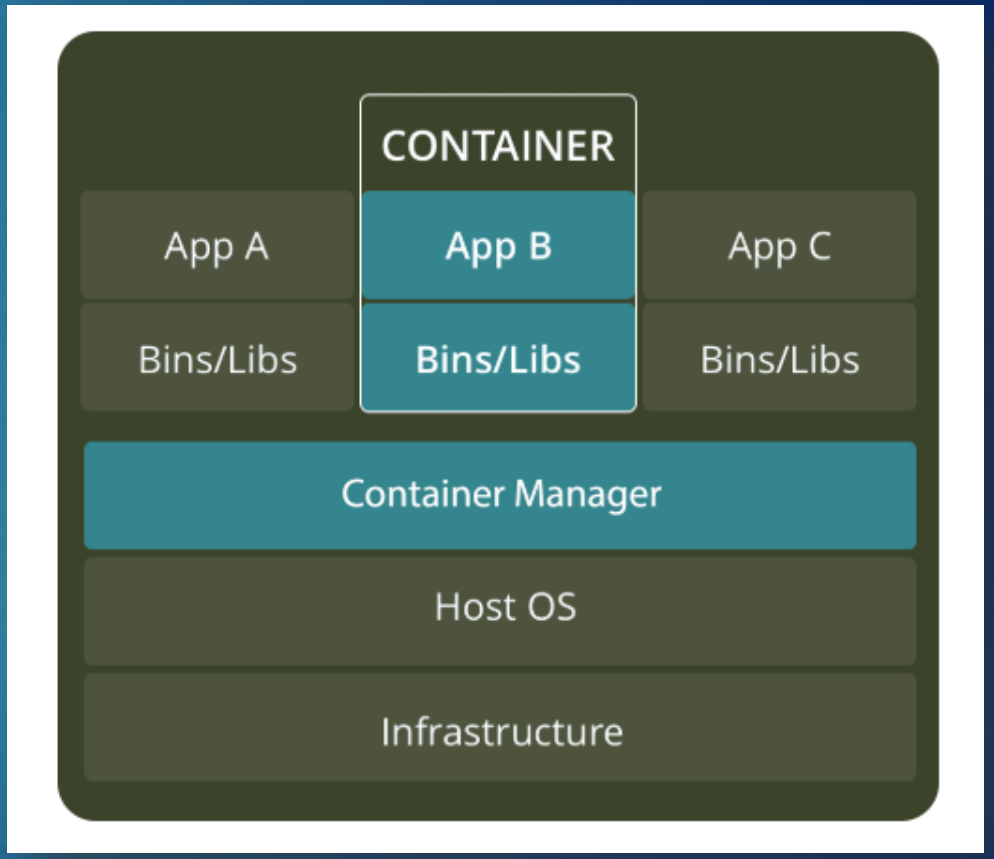
- Answer: A lot
- Ensuring everything works consistently across all platforms is a major challenge
- As a result, these permutations lead to what is referred to as the "Matrix from Hell"
- This is where containers can help



Source: http://pointful.github.io/docker-intro/

Containers - Overview

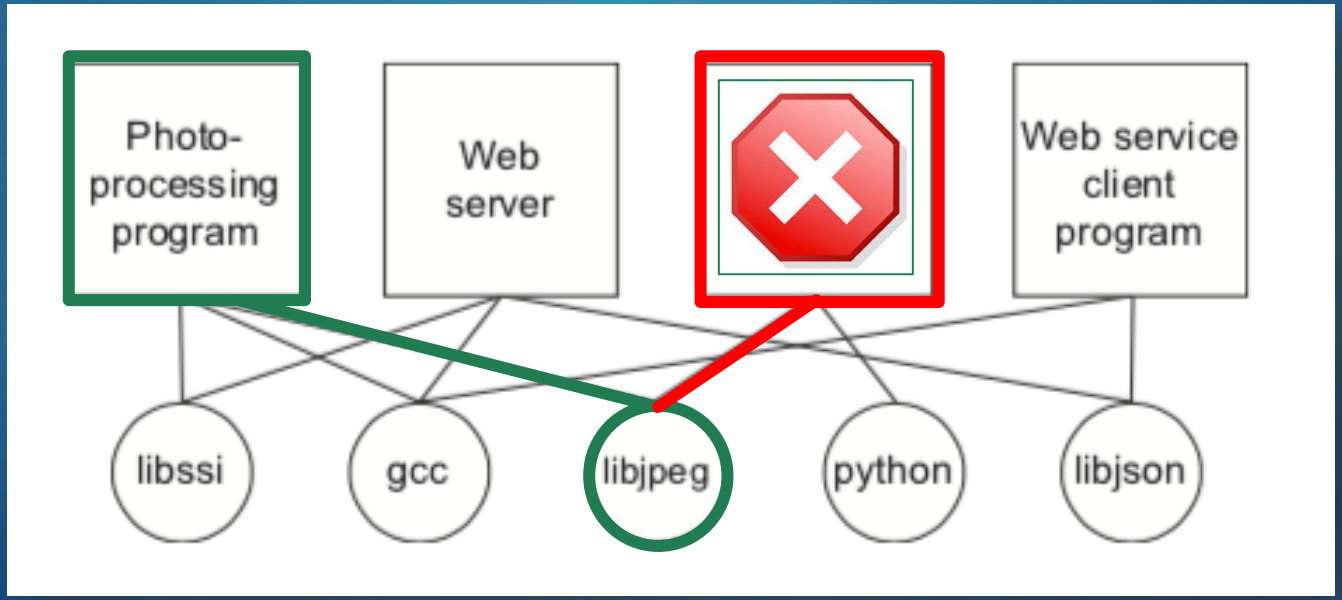
- A container sits on top of a physical server and its host OS – typically Linux or Windows
- Each container shares the host OS kernel and usually the binaries and libraries
- Shared components are read only
- Because they all share a single, common operating system, it is much simpler for things like bug fixes and patches
- Containers (only megabytes in size) use less resources than VMs
- It's possible to fit multiple containers onto a single server



Source: https://www.backblaze.com/blog/vm-vs-containers/

Application Management – Common Challenge

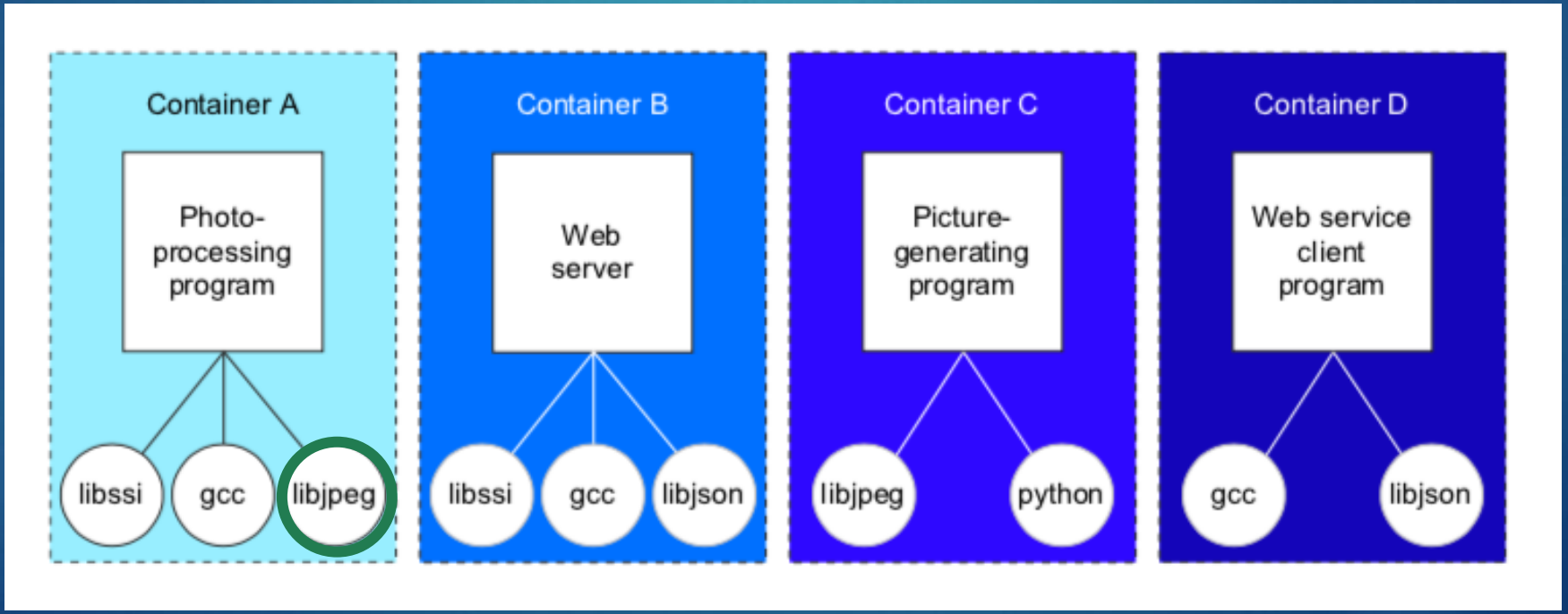
- On a typical server, applications may be dependent on libraries that are shared with other applications
  - Example: They may require exclusive access such as network connection or a file
- Updating one of these dependencies may inadvertently break another application



Source: NICKOLOFF, J. (2019). *DOCKER IN ACTION*. [S.l.]: O'REILLY MEDIA.

Application Management – Solution

- Containerizing applications allows each to run in isolation with impacting other containers
- Dependencies can be safely updated without breaking applications in other containers



Source: NICKOLOFF, J. (2019). *DOCKER IN ACTION*. [S.I.]: O'REILLY MEDIA.

Containers Example – Scenario #1

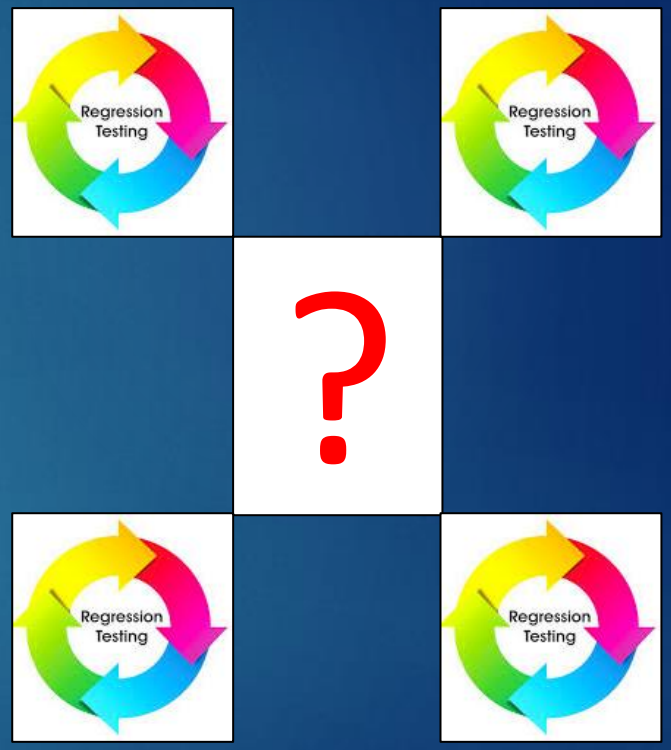
- Problem:
  - You have hundreds of applications running on premise and need to move everything to the cloud
- Challenges:
  - Porting everything to the cloud requires re-compiling and testing on new servers
  - Idle servers can be expensive if they are not fully utilized
  - What if you do not have the original source code to re-compile?

Containers Example – Scenario #1

- Solution:
  - Porting everything to containers first on-premise, test and validate
  - These can move to the cloud with no re-compiling
  - Because of this portability, they could also be easily be moved from one cloud (AWS) another cloud vendor (Azure or Google) and continue to work
  - This avoids vendor lock-in

Containers – Example #2

- Problem:
  - You need to replicate your testing environment for dozens of testers in the cloud
  - The environment, which includes a database, must be configured the same to avoid any testing discrepancies
- Challenges:
  - Creating multiple and isolated test environments can be a time consuming and error prone process
  - The number of servers to support the testing environment can get costly
  - Ensuring the data is exactly same across all systems can add additional complexities

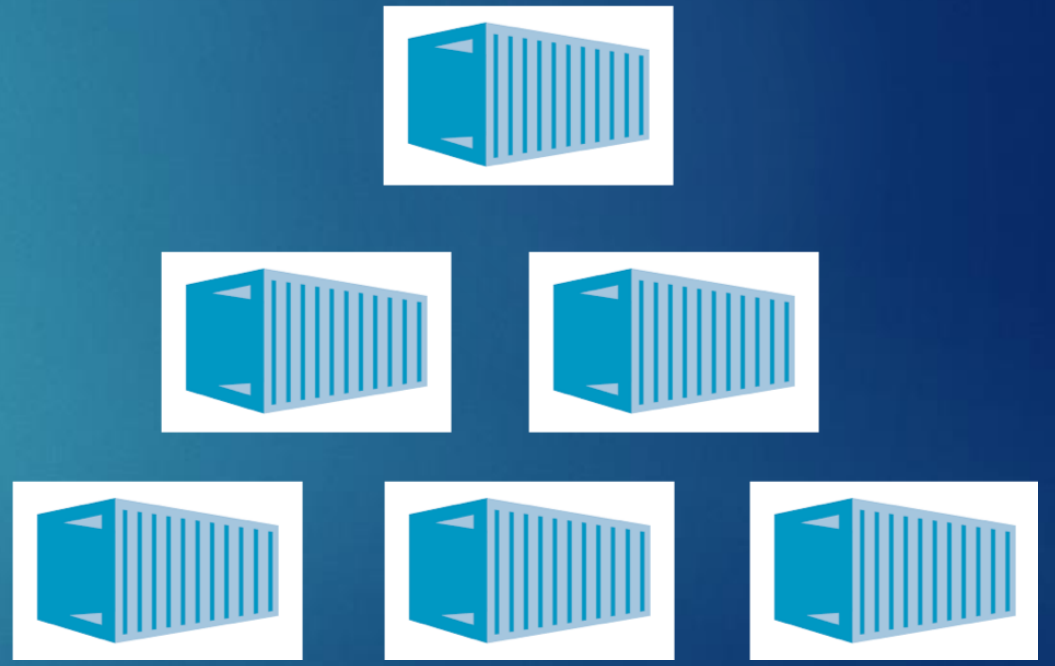


Containers – Example #2

- Solution:
  - The entire testing environment (including database) can be containerized
  - Multiple copies can be spun up on demand and quickly torn down after testing to avoid "cloud waste"
  - This saves time and money and ensures a consistent testing environment

Container Solutions

- Docker is by far the most popular
  - Based on Linux kernel
  - ~84% of the market share
- Other solutions:
  - Podman
  - Containerd
  - Buildah
  - Apache Mesos
  - LXC (Linux Containers)
- For this class, we will be using Docker

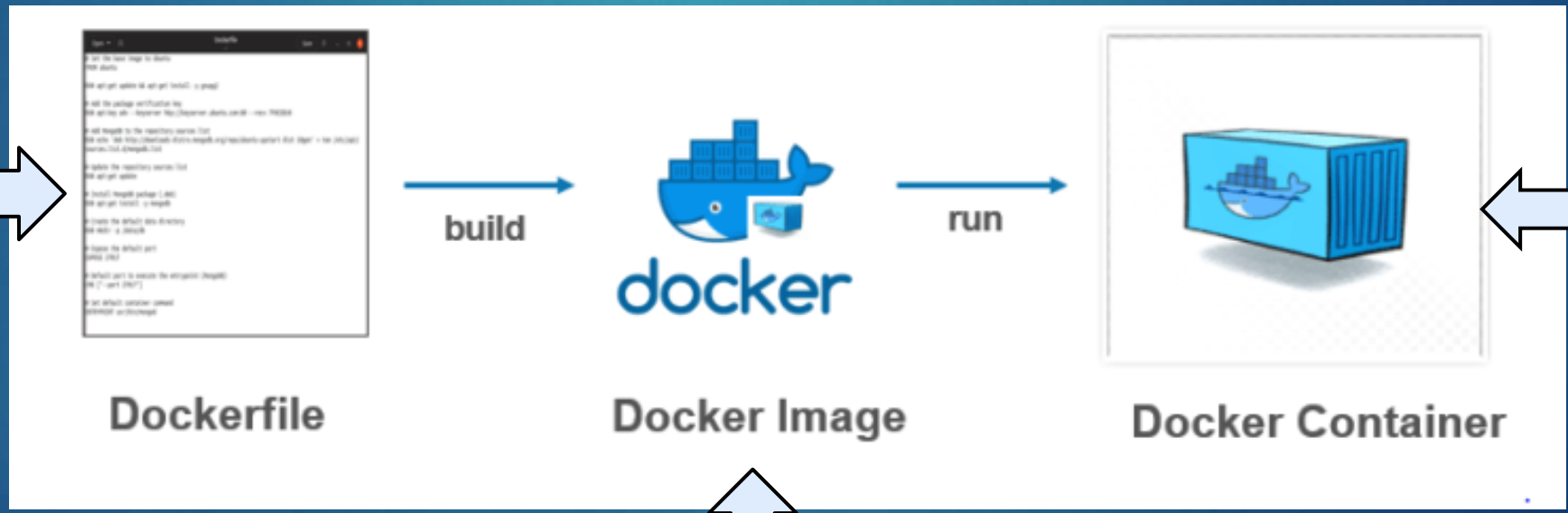


Source: https://6sense.com/tech/containerization

What is Docker?

An open-source containerization engine, which automates the packaging, shipping, and deployment of any software applications that are presented as lightweight, portable, and self-sufficient containers, that will run virtually anywhere

A **Dockerfile** is a text document that contains all the commands a user could call on the command line to assemble an image

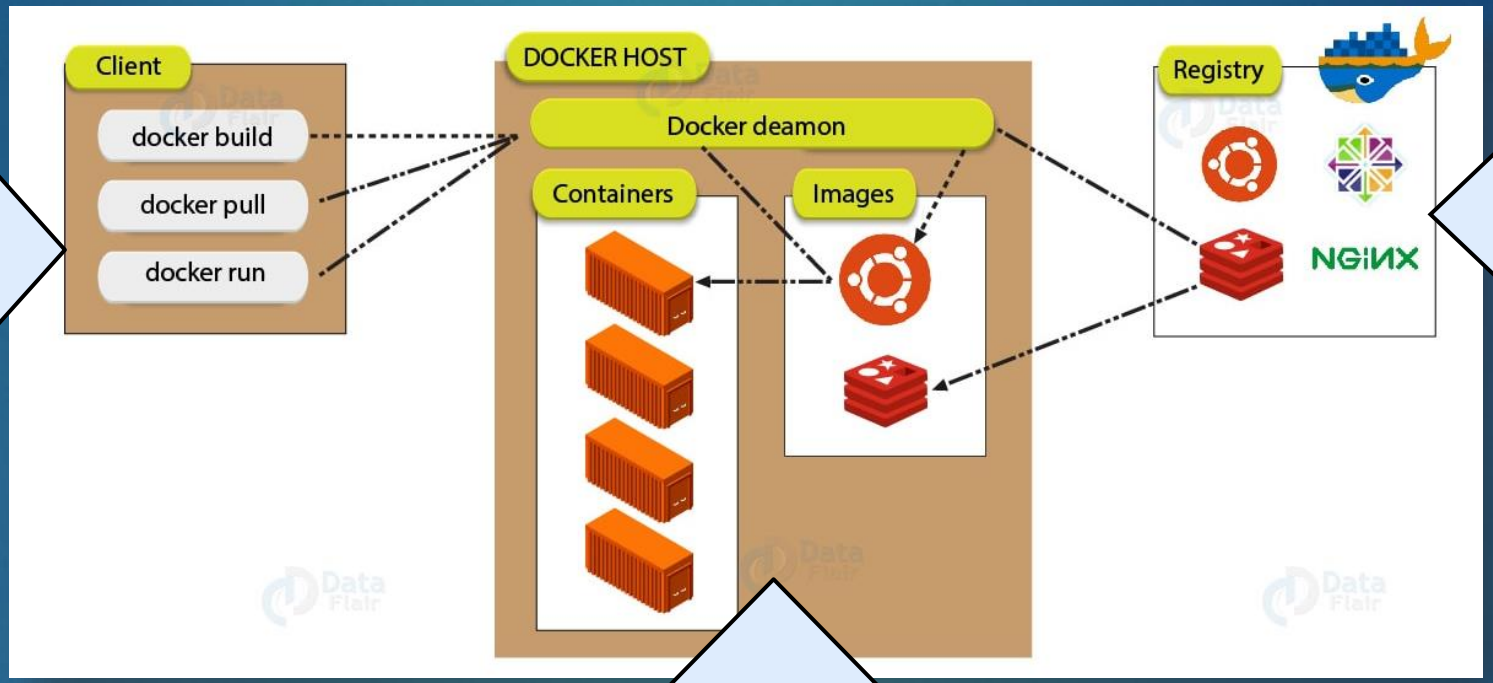


A **Docker container** is a software bucket comprising everything necessary to the software independently

A **Docker Image** is a file used to execute code in a Docker container

How does Docker Work?

- **Docker Client** is the major way that provides communication between many dockers users to Docker
- It sends the commands (e.g. docker build, docker run, etc.) to the **Docker Daemon**, which carries them out



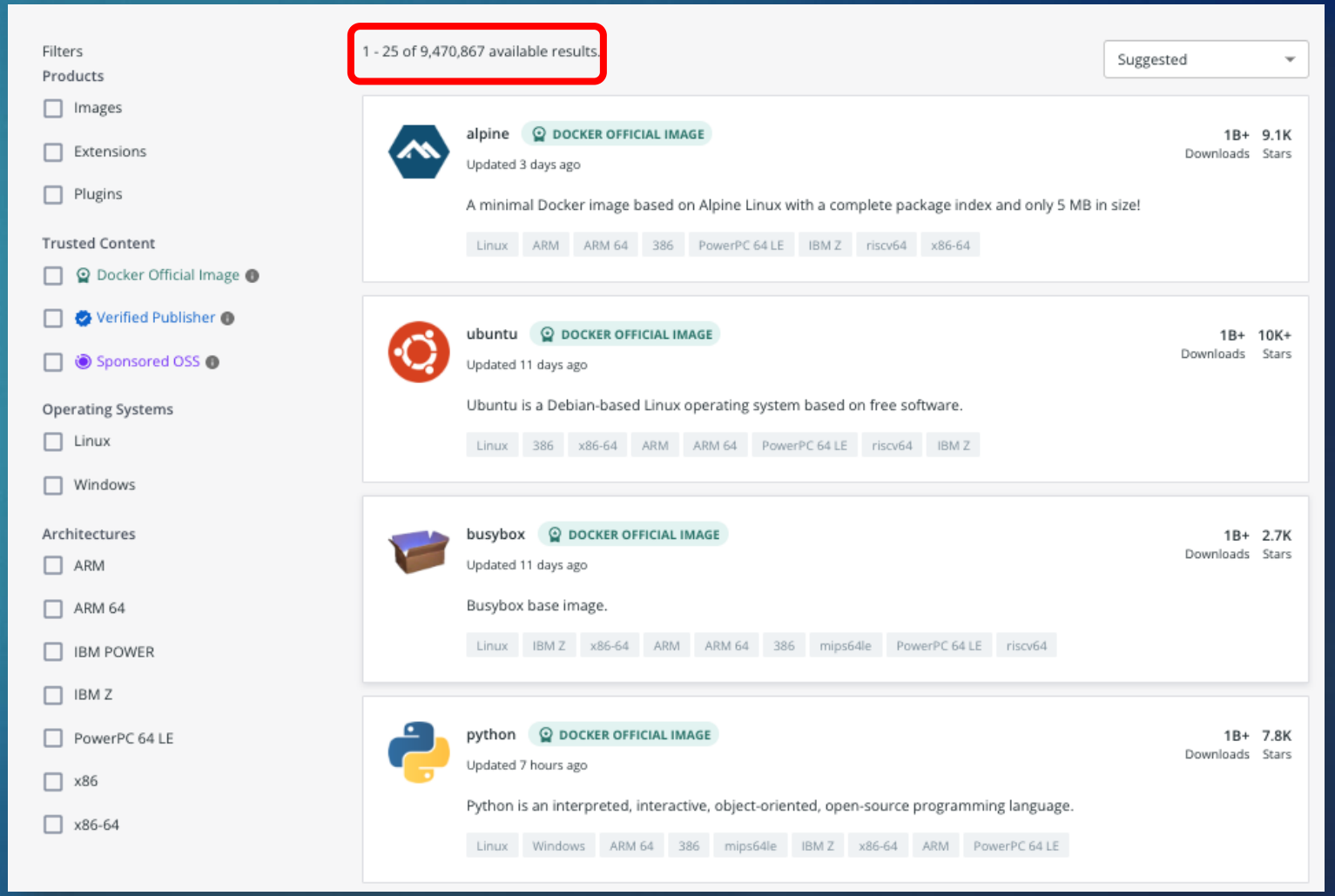
- **Docker Registry** is the where Docker images are stored
- **Docker Hub** is a public registry that anyone can access and configure

- **Docker Host** runs the **Docker Daemon**.
- The Docker Daemon listens for Docker API requests such as 'docker run', 'docker builds'
- It also manages Docker objects like images, containers, networks, etc.

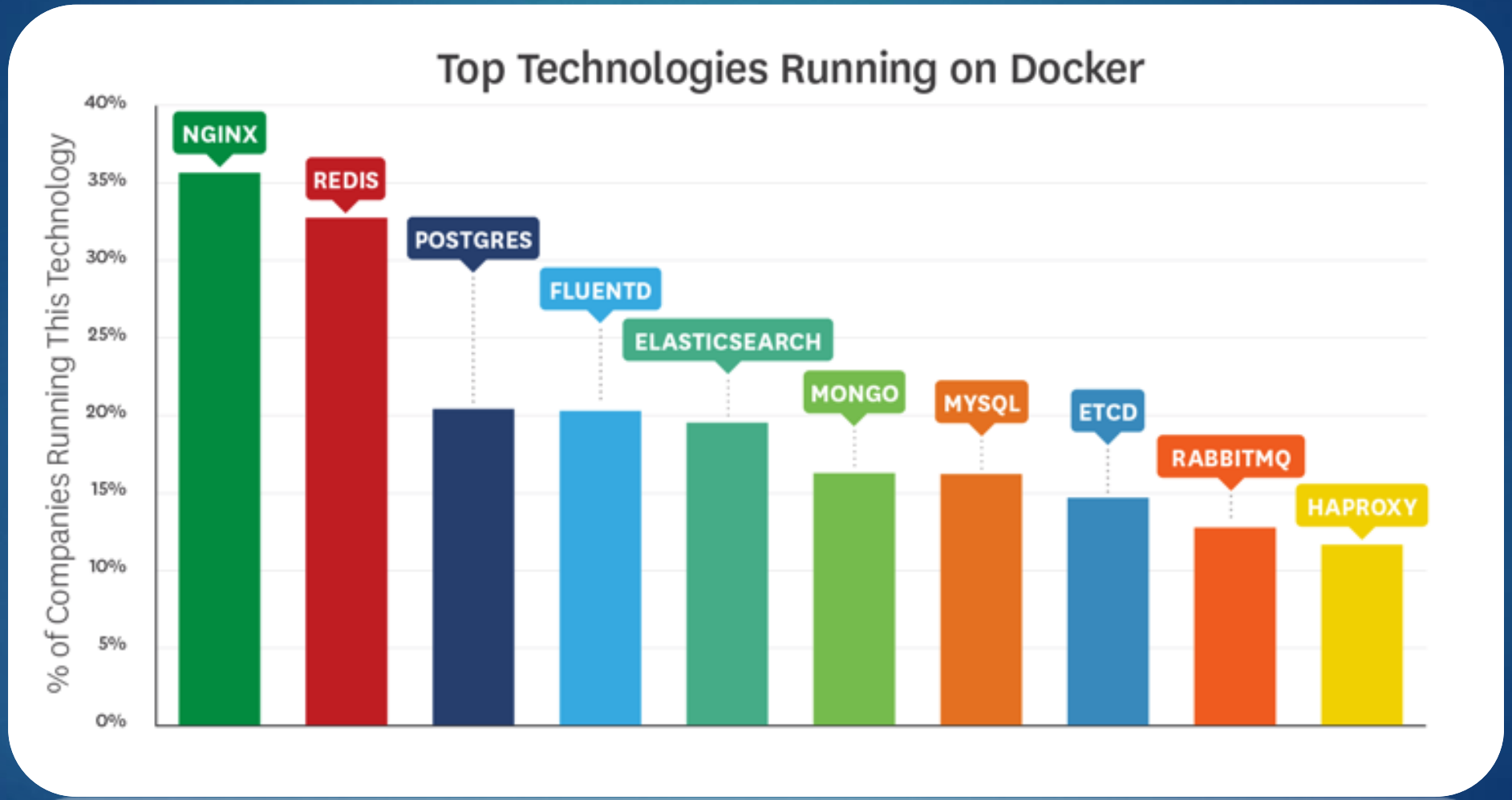
Source: https://data-flair.training/blogs/docker-architecture/

Docker Hub (hub.docker.com) – Repository

- Service provided by Docker for finding and sharing container images
- There is over ~9 million public images available
  - ~100,000 images that are vendor provided
- Some are fully functional; others can provide a base (e.g. Python) that you can then customize



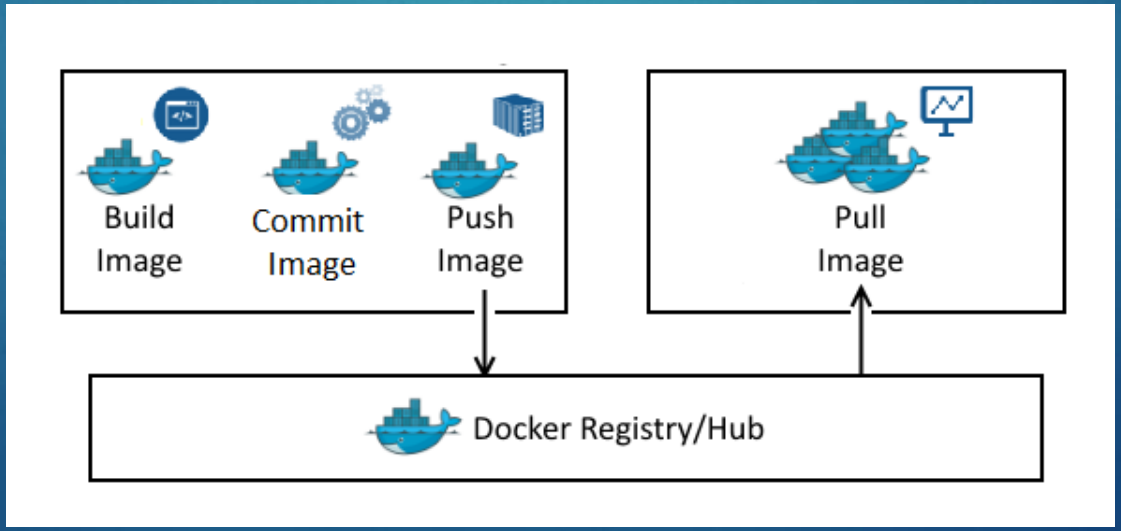
Most widely used images for Docker



Source: https://www.datadoghq.com/docker-adoption/

Creating your own Docker Image

- There are a couple ways you can create your own image
  1. Download a base image (e.g. Python) and connect to it via bash shell to customize your own programs and save as a new image
  2. Create your own image from scratch using Dockerfile to create an image
- When you are done, you can publish to a registry such as Docker Hub
- You can then pull the image to another machine running Docker



Docker - Demo

1. Pull a pre-built Docker nginx web server image from DockerHub to your local system

```
$ docker pull nginx
```

2. Run the Docker container

```
$ docker run -d -p 8080:80 nginx
```

- This starts the nginx container in detached mode (-d), mapping port 80 in the container to port 8080 on your machine (http://localhost:8080)

3. Stop the running Container

```
$ docker ps
$ docker stop <container-id>
```

- Use docker ps to list running containers and docker stop to stop the container by its ID

Docker & DockerHub - Demo

1. This Dockerfile uses the official nginx image and replaces the default web page with a custom index.html

Dockerfile

```
# Dockerfile
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
```

index.html

```
<!-- index.html -->
<html>
<head><title>Welcome to My Custom NGINX!</title></head>
<body><h1>Hello, class! This is my custom NGINX page.</h1></body>
</html>
```

2. Build the image, tagging it as **custom-nginx**

```
docker build -t custom-nginx .
```

3. Run it!

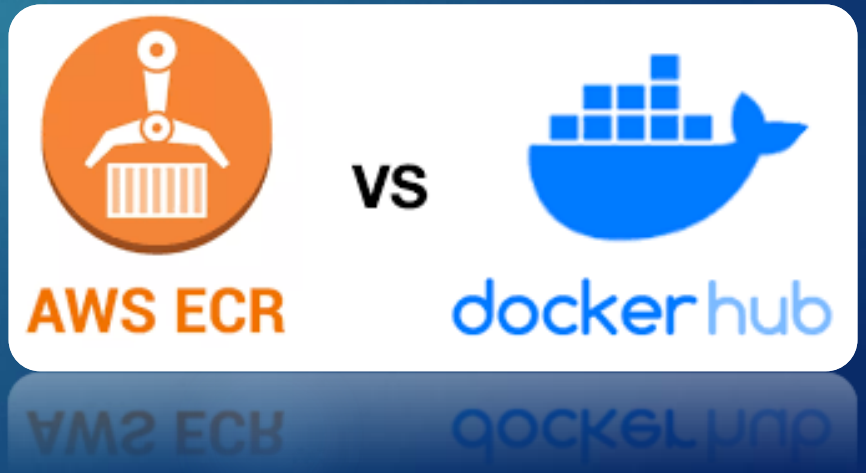
```
docker run -d -p 8080:80 custom-nginx
```

4. Tag the custom image with your DockerHub username and uploads it to your repository on DockerHub

```
$ docker tag custom-nginx <your-dockerhub-username>/custom-nginx
$ docker push <your-dockerhub-username>/custom-nginx
```

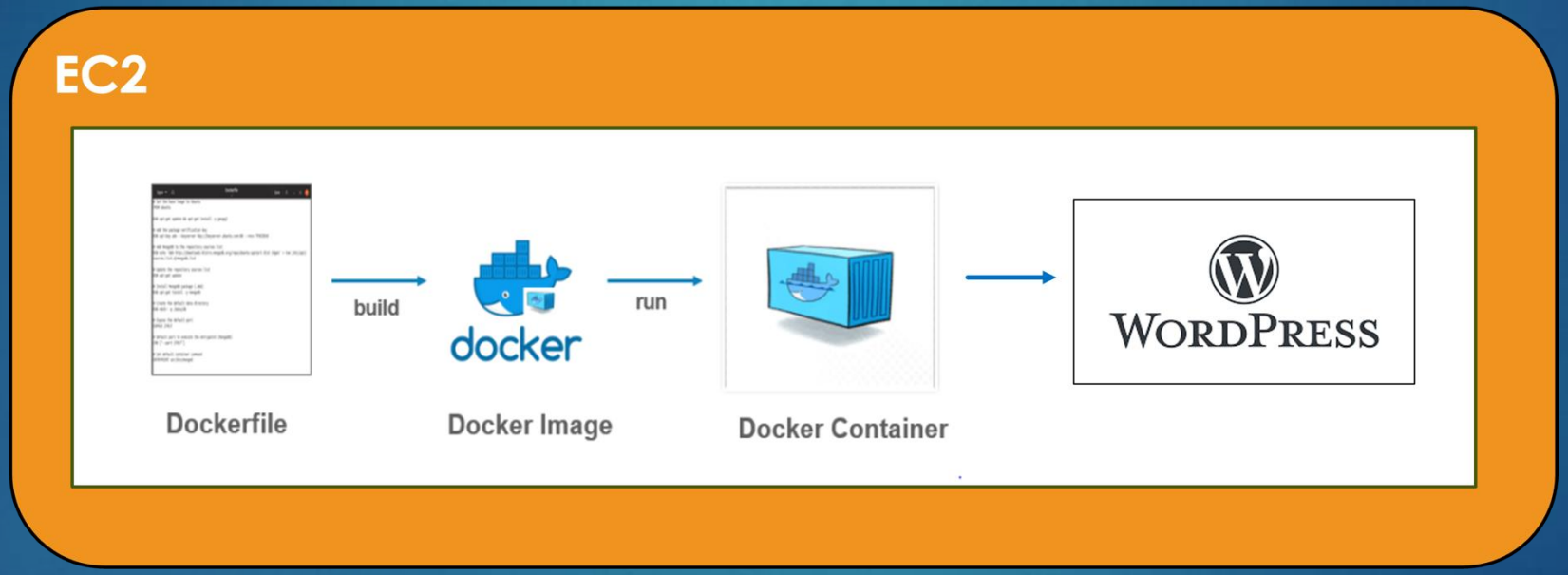
Amazon Elastic Container Registry (ECR)

- Fully managed container registry by AWS, enabling secure storage, management, and deployment of Docker container images
- Provides image encryption at rest, IAM-based access controls, and vulnerability scanning with Amazon Inspector
- DockerHub vs. ECR:
  - DockerHub has public repositories, ECR repositories are private by default, enhancing security
  - DockerHub is a globally accessible service while ECR operates within AWS regions, ensuring low-latency access
  - DockerHub's requires on a standalone account, ECR uses AWS credentials for authentication



Containerization Activity

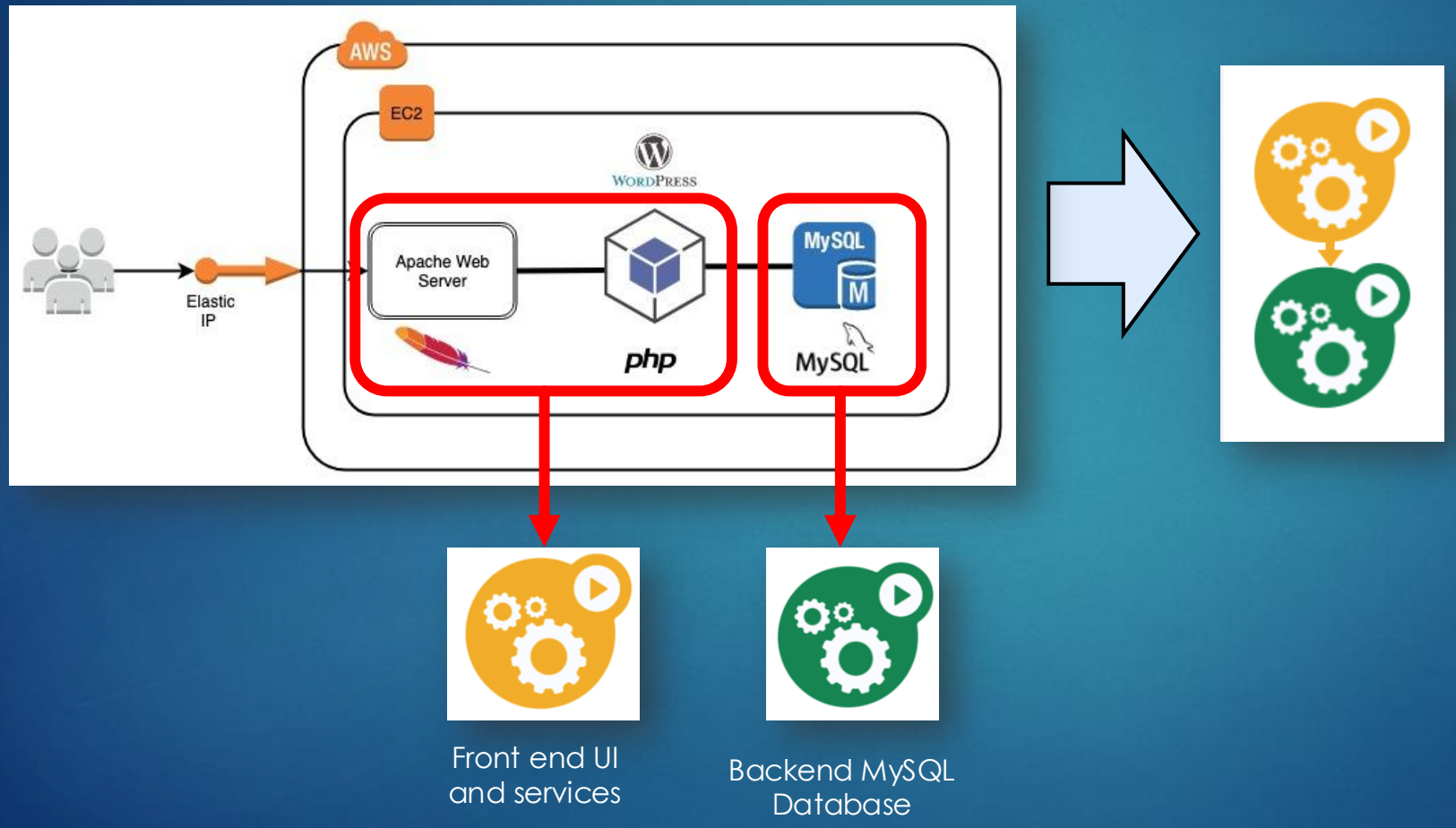
- Going back to the WordPress on Docker activity, what are some of the limitations with this solution?



- How would you make this more scalable and resilient?

Containerization Activity

- A good first step is to split up the functionality into separate containers



Good:

- We have containerized our application into 2 parts
- We can update either one independently (e.g. switch database)

Bad:

- Single point of failure for each container
- Not scalable

Containerization Activity

- Next, we can duplicate the containers to address any concerns with scalability

Good:

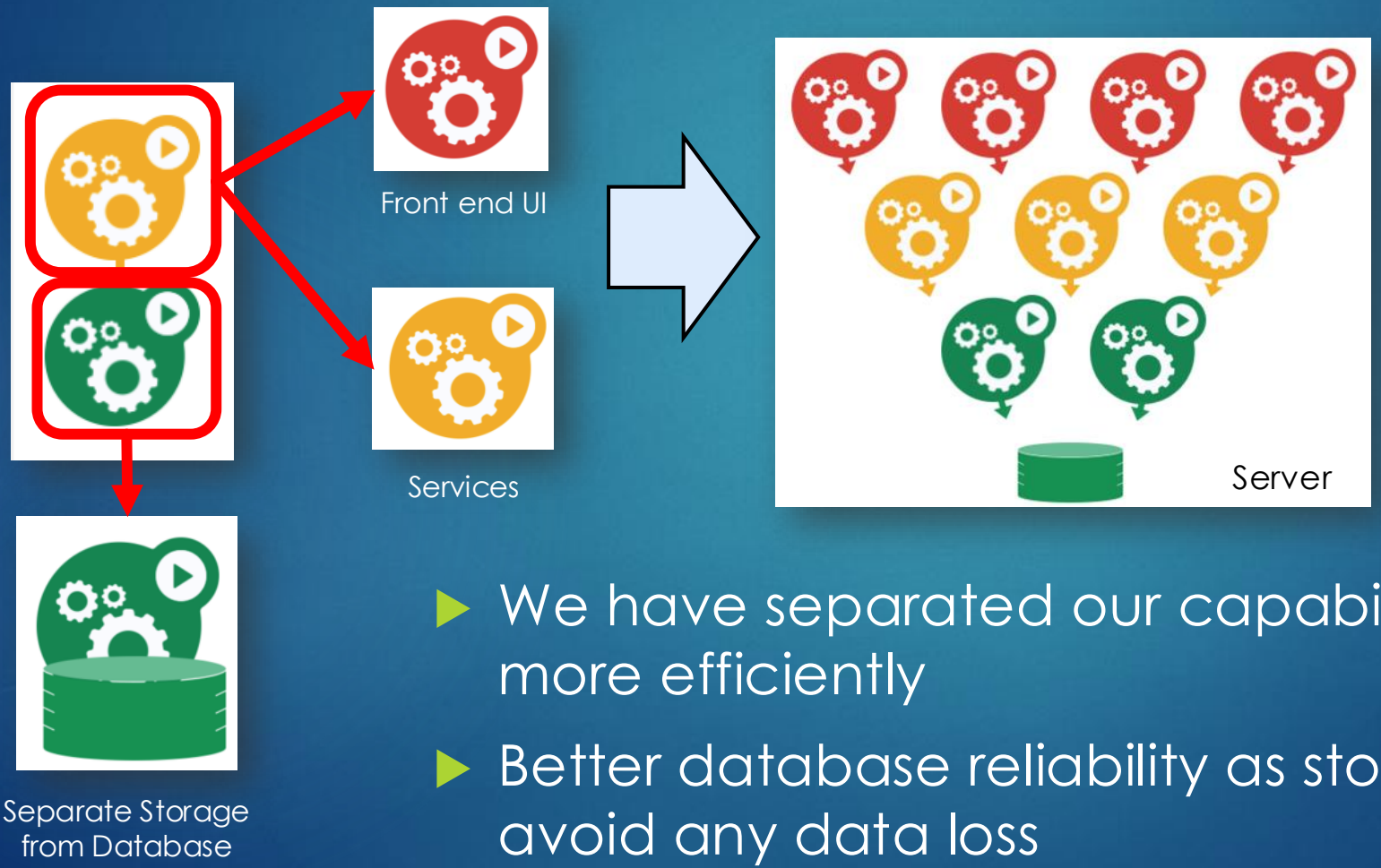
- We have a more scalable front end
- We can update some containers independently with no downtime

Bad:

- Front end has too many responsibilities (UI and services)
- Database goes down, we lost everything
- Duplicating database doesn't help

Containerization Activity

- Split up the front-end services to separate responsibilities



- We have separated our capabilities and can scale more efficiently
- Better database reliability as storage is detached to avoid any data loss

Containerizing WordPress

- For this activity, you will be using your WordPress container from the previous activity and separating into 2 containers: WordPress UI and MySQL database
- The new images will be stored in the AWS Elastic Container Registry (ECR) in preparation for the final Containerization activity
- This activity can be accessed at Assignments > Activity – WordPress and MySQL on Docker
- Note: the next activity will require this activity to be successfully completed